

HPCC场景下使用Volcano

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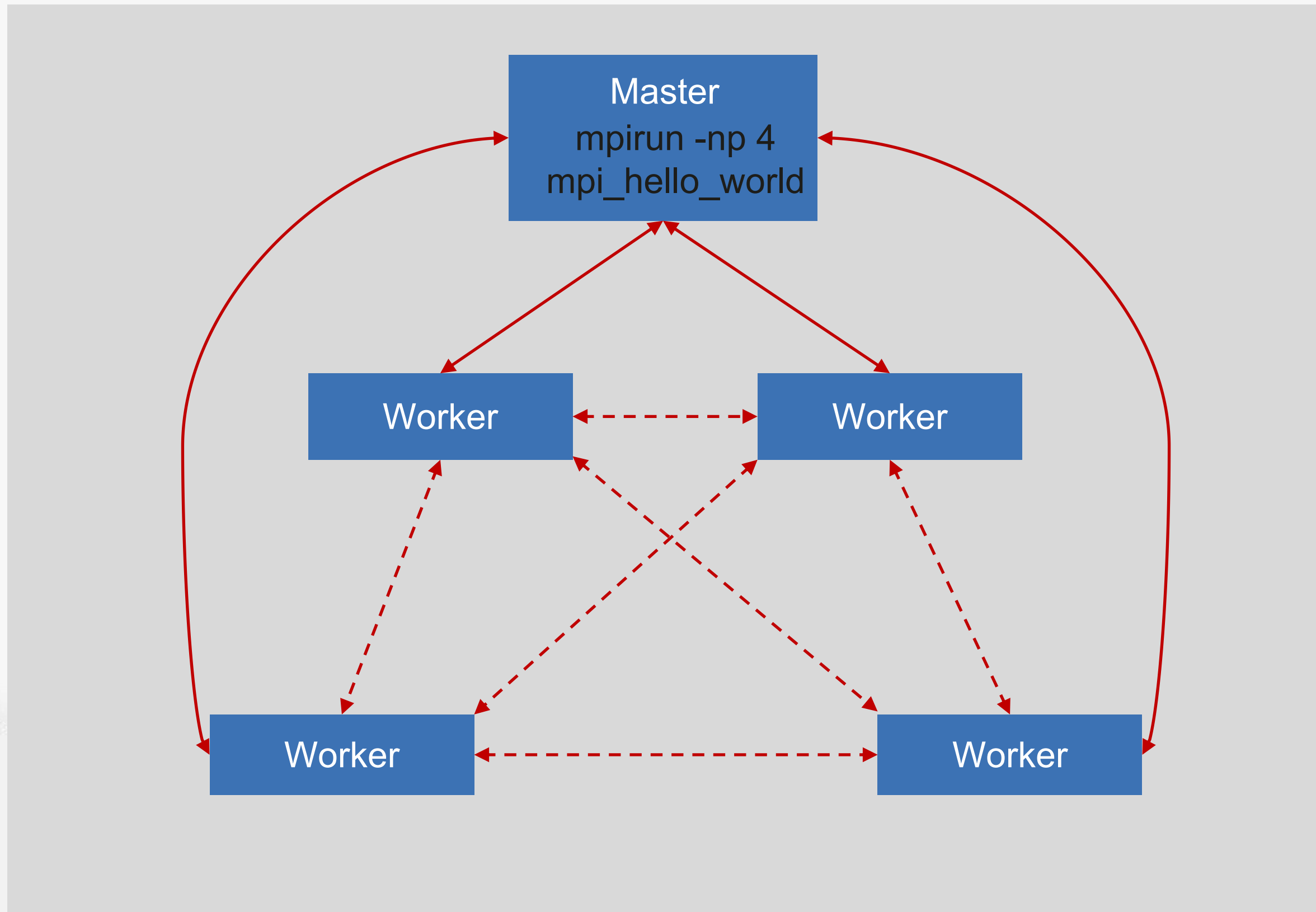
01

Volcano运行HPC作业



Volcano运行HPC作业

在容器中运行MPI作业:



Volcano支持MPI作业关键技术:

- 支持定义多个Pod模板
- 支持Gang调度能力
- Master/Worker容器中支持主机IP映射（通过kubernetes headless service）
- 使用SSH实现主容器和工作容器之间的免密码登录
- MPI作业生命周期管理

02

演示：运行MPI作业

作业配置

metadata:

name: mpi-job

labels:

"volcano.sh/job-type": "MPI"



根据业务需求设置作业类型

spec:

minAvailable: 3

schedulerName: volcano

plugins:

ssh: []



开启SSH免密码认证

svc: []



提供网络信息、主机IP映射文件及K8S headless service

policies:

- event: PodEvicted

action: RestartJob



配置作业重启策略



作业配置

tasks:

- replicas: 1
name: mpimaster

policies:

- event: TaskCompleted
action: CompleteJob

template:

spec:

containers:

- command:

- /bin/sh

- -c

- |

MPI_HOST=`cat /etc/volcano/mpiworker.host | tr "\n" ","`;
mkdir -p /var/run/sshd; /usr/sbin/sshd;
mpirun --allow-run-as-root --host \${MPI_HOST} -np 2

mpi_hello_world;

image: volcanosh/example-mpi:0.0.1

name: mpimaster

ports:

- containerPort: 22

name: mpijob-port

配置作业终止策略。当mpirun完成时，整个MPI作业被视为完成。

Master容器启动命令。主机IP映射文件存储在 **/etc/volcano** 中。首先启动ssh服务，然后运行mpirun启动mpi作业。

暴露22端口



作业配置

```
tasks:
  - replicas: 2
    name: mpiworker
    template:
      spec:
        containers:
          - command:
            - /bin/sh
            - -C
            - |
              mkdir -p /var/run/sshd; /usr/sbin/sshd;
            image: volcanosh/example-mpi:0.0.1
            name: mpimaster
            ports:
              - containerPort: 22
                name: mpijob-port
```



Worker容器启动命令。仅启动sshd服务



暴露22端口

演示

MPI作业状态:

NAME	READY	STATUS	RESTARTS	AGE
mpi-job-mpimaster-0	1/1	Running	1	22s
mpi-job-mpiworker-0	1/1	Running	0	22s
mpi-job-mpiworker-1	1/1	Running	0	22s

NAME	READY	STATUS	RESTARTS	AGE
mpi-job-mpimaster-0	0/1	Completed	1	2m12s



MPI作业运行中
Master pod可能会因网络建设延迟而在运行过程中重启



MPI作业结束
作业结束后保留Master Pod，删除Worker Pod

MPI作业结果:

Warning: Permanently added 'mpi-job-mpiworker-0.mpi-job,10.0.0.157' (ECOSA) to the list of known hosts.
Warning: Permanently added 'mpi-job-mpiworker-1.mpi-job,10.0.0.159' (ECOSA) to the list of known hosts.

Hello world from processor mpi-job-mpiworker-0, rank 0 out of 2 processors
Hello world from processor mpi-job-mpiworker-1, rank 1 out of 2 processors



警告信息



作业输出



演示

```
[root@volcano-demo mpi-demo]$
```

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03

演示：运行Argo作业

Argo介绍

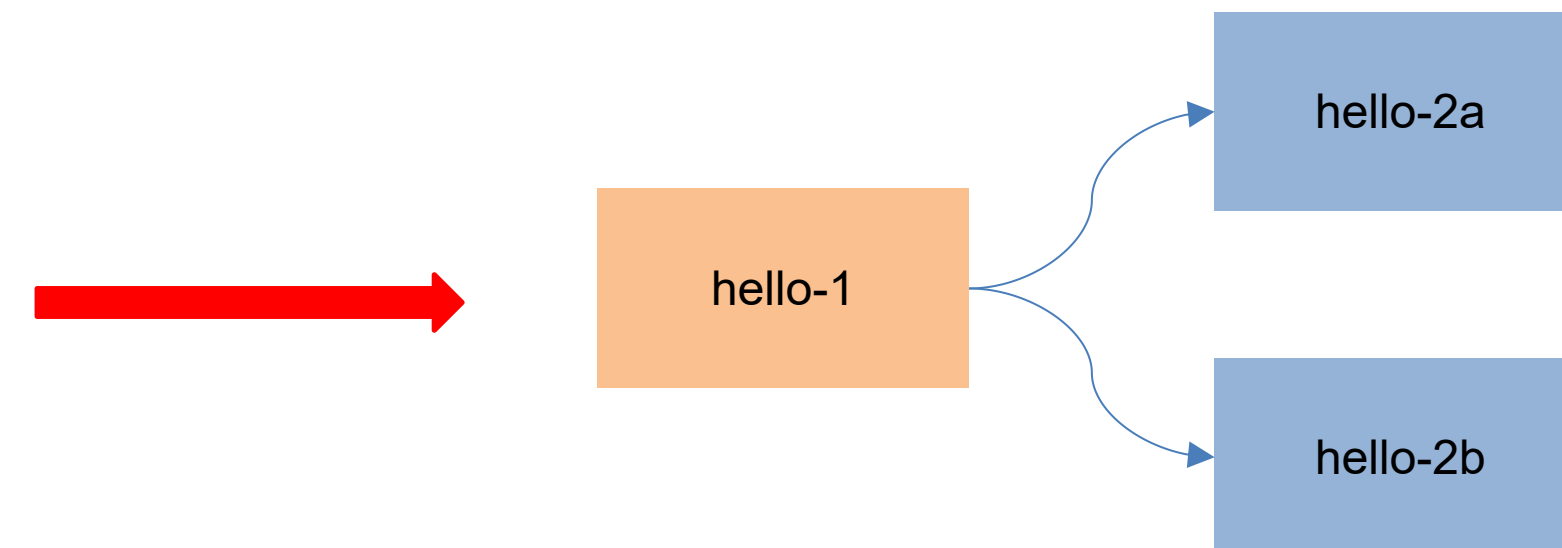
Argo Workflows是一个开源项目，通过Kubernetes CRD为Kubernetes提供容器原生 workflow 支持。

- 工作流的每一步都是一个容器
- 将多步 workflow 建模为一系列任务，使用有向无环图（DAG）描述任务之间的依赖关系
- 可以在短时间内轻松运行用于机器学习或数据处理的计算密集型作业
- 在Kubernetes上运行CI/CD管道，无需复杂的软件配置



定义 workflow

```
spec:
  entrypoint: volcano-step-job
  serviceAccountName: argo
  templates:
  - name: volcano-step-job
    steps:
    - - name: hello-1
      template: hello-tmpl
      arguments:
        parameters: [{name: message, value: hello1}, {name: task,
value: hello1}]
    - - name: hello-2a
      template: hello-tmpl
      arguments:
        parameters: [{name: message, value: hello2a}, {name: task,
value: hello2a}]
    - name: hello-2b
      template: hello-tmpl
      arguments:
        parameters: [{name: message, value: hello2b}, {name: task,
value: hello2b}]
```



创建Volcano作业

```
- name: hello-tmpl
...
resource:
.....
manifest: |
  apiVersion: batch.volcano.sh/v1alpha1
  kind: Job
  metadata:
    generateName: step-job-{{inputs.parameters.task}}-
    ownerReferences:
      - apiVersion: argoproj.io/v1alpha1
        blockOwnerDeletion: true
        kind: Workflow
        name: "{{workflow.name}}"
        uid: "{{workflow.uid}}"
  spec:
    minAvailable: 1
    schedulerName: volcano
    policies:
      - event: PodEvicted
        action: RestartJob
    plugins:
      ssh: []
      env: []
      svc: []
    maxRetry: 1
    queue: default
    tasks:
      - replicas: 2
        name: "default-hello"
        template:
          metadata:
            name: helloworld
          spec:
            containers:
              - image: docker/whalesay
                imagePullPolicy: IfNotPresent
                command: [cowsay]
                args: ["{{inputs.parameters.message}}"]
                name: hello
            resources:
              requests:
                cpu: "100m"
            restartPolicy: OnFailure
```



将 workflow 中的步骤定义成 volcano 作业



演示

```
[root@volcano-demo argo-demo]$ # check argo controller status
```

```
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```


加入Volcano



Website: <https://volcano.sh/en/>



Github: <https://github.com/volcano-sh/volcano>



Slack Channel: <https://volcano-sh.slack.com/>



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